VII SEMESTER (OPEN ELECTIVE III)	L	T	P	C
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20CE7O01 - SOLID WASTE MANAGEMENT				

COURSE OUTCOMES:

After the completion of the course, student will be able to

CO1: Recall classification of solid waste generated.
CO2: Know the collection systems of solid waste of a town.
CO3: Analyze the importance of transfer and transport of solid waste.
CO4: Apply the knowledge in processing of solid waste.
CO5: Design treatment of municipal solid waste and landfill.

UNIT- I:
Introduction to Solid Waste Management:
Goals and objectives of solid waste management, Classification of Solid Waste – Factors Influencing generation of solid waste - sampling and characterization – Future changes in waste composition, major legislation, monitoring responsibilities.

UNIT- II:
Collection of Solid Waste:
Type and methods of waste collection systems, analysis of collection system optimization of collection routes– alternative techniques for collection system.

UNIT- III:
Transfer and Transport:
Need for transfer operation, compaction of solid waste - transport means and methods, transfer station types and design requirements.

UNIT- IV:
Processing and Treatment:
Processing of solid waste – Waste transformation through combustion and composting, anaerobic methods for materials recovery and treatment – Energy recovery – biogas generation and cleaning– Incinerators.

UNIT- V:
Disposal of Solid Waste:
Methods of Disposal, Landfills: Site selection, design and operation, drainage and leachate collection systems –designated waste landfill remediation.

Text/ Reference books:

1. George Techobanoglous, Frank Kreith , Integrated Solid Waste Management- McGraw Hill Publication, 1993.
2. R.Saravanan, R.Dinesh Kumar, A.Suriya , Muncipal solid waste management, Lakshmi publications-2015.
3. Vesilind, P.A., Worrell, W., Reinhart, D.,“Solid Waste Engineering”, Cenage learning, New Delhi, 2004.